

DS Unit 4 — Graphs & Graph Algorithms Complete Notes

Data Structures (DS) · Unit 4 · Graphs & Graph Algorithms · 2024

Overview

These notes cover **Graphs & Graph Algorithms**. Core data structures and the algorithms built on them — arrays, lists, stacks, trees, graphs, sorting and hashing. Each section states the core idea, a worked intuition and the points examiners look for.

1. Graph Representations

Adjacency matrices use $O(V^2)$ space while adjacency lists use $O(V + E)$.

Why it matters: Graph Representations appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of graph representations in one line.
- A small example showing graph representations in practice.
- Common mistakes students make and how to avoid them.

2. Breadth-First Search

BFS explores level by level using a queue and finds shortest paths in unweighted graphs.

Why it matters: Breadth-First Search appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of breadth-first search in one line.
- A small example showing breadth-first search in practice.
- Common mistakes students make and how to avoid them.

3. Depth-First Search

DFS explores as deep as possible using recursion or a stack and underlies topological sort.

Why it matters: Depth-First Search appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of depth-first search in one line.
- A small example showing depth-first search in practice.
- Common mistakes students make and how to avoid them.

4. Minimum Spanning Trees

Prim's and Kruskal's algorithms find a spanning tree of minimum total weight.

Why it matters: Minimum Spanning Trees appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of minimum spanning trees in one line.
- A small example showing minimum spanning trees in practice.
- Common mistakes students make and how to avoid them.

5. Dijkstra's Algorithm

Dijkstra computes single-source shortest paths for non-negative edge weights using a priority queue.

Why it matters: Dijkstra's Algorithm appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of dijkstra's algorithm in one line.
- A small example showing dijkstra's algorithm in practice.
- Common mistakes students make and how to avoid them.

6. Topological Sorting

A topological order of a DAG lists each vertex before all vertices it points to.

Why it matters: Topological Sorting appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of topological sorting in one line.
- A small example showing topological sorting in practice.
- Common mistakes students make and how to avoid them.

Quick revision

Graph Representations · Breadth-First Search · Depth-First Search · Minimum Spanning Trees · Dijkstra's Algorithm · Topological Sorting